

Select the most appropriate answer(s) for each of the following questions. Put a circle around your choice(s):

- 1- The following data platform is an example of SQL databases:
☒ a- PostgreSQL
b- Web Tier
c- Neo4J
d- MongoDB
- 2- The key concept behind distributed applications systems is that they are software applications to ensure there is a web tier for system administrators:
a- True
☒ b- False
- 3- In relational database management systems, SQL is used to work and interact with the system. Which statement could be used to retrieve patient record given his age and location?
a- Insert...as...Select...From
☒ b- Select...From...Where
c- Update...Where
d- Delete...Where
- 4- Two of the very key motivations to build distributed systems are (select two):
a- Scalability to reduce down time
☒ b- Availability to reduce down time
☒ c- Scalability to cater for workload growth
d- Availability to cater for workload growth
- 5- TCP/IP sockets is used as the standard communication interface for distributed applications runs beneath different technologies like JDBC, JMS, and REST APIs.
☒ a- True
b- False
- 6- Distributed applications can be built in different topologies with key feature to enable an application to run over different physical or virtualized nodes. All the following are examples of such architecture except for one:
a- N-Tier Applications
b- Peer-to-Peer Applications
☒ c- Node.js
d- Microservices

- 7- The main idea behind using public cloud is to pay as you utilize the services instead of buying and owning the infrastructure components needed for those services.
- ☒ a- True
 - b- False
- 8- The most important tier in Multi-Tier applications where the patient, hospitals, diagnostics, and other data are physically stored is:
- a- EMR
 - b- EPR
 - ☒ c- Database
 - d- AWS
- 9- When modeling the database for an EMR application, you wanted to ensure all the details of the data to be captured and stored is represented clearly. Which tool should be used:
- ☒ a- SQL
 - ☒ b- ERD
 - c- MongoDB
 - d- NoSQL
- 10- Amazon Web Services (AWS) is a collection of components including infrastructure, applications, databases, web servers among others offered in highly available, manageable, and integrated services.
- ☒ a- True
 - b- False
- 11- When architecting distributed applications, two types of requirements are applied, Functional and Non-Functional requirements. Which of the following is essential one of the Functional Requirements?
- a- Scalability
 - ☒ b- User Interface
 - c- Reliability
 - d- Availability
- 12- The core protocol for enabling modern distributed applications to be designed and implemented including different patterns like client/server, n-tier and p2p is:
- a- Microwave
 - ☒ b- TCP/IP
 - c- ADSL
 - d- ATM
- 13- The different functionality of applications when it comes to the nature of processing are:
- ☒ a- OLTP, OLAP and MDM
 - b- OLTP, OLAP and Microservices
 - c- OLAP, OLTP and N-Tier
 - d- None of the above
- 14- ATM machines is one example of fat-client applications in the Client-Server architectural model.
- ☒ a- True
 - b- False

- 15- In relational database management systems, a transaction is defined as:
- a- A collection of operations that performs an action on one table
 - ☒ b- A collection of operations that performs a single logical function
 - c- A collection of operations that performs an action on two or more databases
 - d- A collection of operations that performs a single database statement
- 16- Distributed applications architecture includes four layers. Following are three of them (select three):
- ☒ a- Presentation
 - ☒ b- Data Handling
 - c- Transactions
 - ☒ d- Database
- 17- In distributed component architecture, the component that lays the central role in enabling other components to communicate together:
- a- Web Server
 - b- Client
 - ☒ c- Middleware
 - d- None of the above
- 18- BitTorrent and Bitcoin network are examples of the following distributed applications architecture:
- a- Multi-Tier Client-Server
 - b- Client-Server
 - ☒ c- Peer-to-Peer
 - d- Thin Client
- 19- Software as a service (SaaS) involves hosting the software remotely and providing access to it over the Internet:
- ☒ a- True
 - b- False
- 20- In distributed applications, two models of communication can be utilized which are Synchronous and Asynchronous. Which of the following are attributes of the Asynchronous model (select all that apply)?
- ☒ a- Sender and receiver don't have to be connected at the same time
 - b- Sender and receiver must be connected at the same time
 - c- Sender and receiver must use the same programming and database technologies
 - ☒ d- Sender and receiver can exchange multiple messages on different topics
 - ☒ e- Sender and receiver can use different sockets on different nodes
- 21- Electronic Health Records (EHR) applications belongs to the which type of processing?
- a- OLTP
 - b- OLAP
 - ☒ c- MDM
 - d- None of the above
- 22- "Presenting each user of the software with the impression that they are working with their own copy of the system while, at the same time, making efficient use of system resources" called:
- a- Availability
 - ☒ b- Multi-tenancy
 - c- Service Configuration

- 23- Electronic Health Records (EHR) applications' main role is to provide replacement for other medical applications like HMIS, PACS:
- a- True
 - ☒ b- False
- 24- Amazon Web Services (AWS) provides a way to manage relational databases on public cloud with cost effective way including running multiple DB instance. This service name is:
- a- AWS S3
 - ☒ b- AWS RDS
 - c- AWS Compute
 - d- None of the above
 - e- All of the above
- 25- The row and column model of representing and storing data is used for the following system:
- a- Key-value Stores
 - ☒ b- MySQL
 - c- Web Servers
 - d- Graphs
- 26- Which of the following is an element of relational model for database management systems?
- a- Table
 - b- Row
 - c- Column
 - ☒ d- All of the above
- 27- The architecture of EHR doesn't require a database tier as it just connects to other medical applications:
- a- True
 - ☒ b- False
- 28- In the API-First Architecture, only REST APIs is included in the distributed application architecture with no need for any other tiers or services:
- a- True
 - ☒ b- False
- 29- A database system architecture like MySQL contains only physical data storage on disks hence its role is to provide disk management not memory management:
- a- True
 - ☒ b- False
- 30- When designing a database schema, ERD diagrams are used where the entities are representing:
- a- Rows
 - b- Columns
 - ☒ c- Tables
 - d- Document
 - e- None of the above

End of the Mid Term